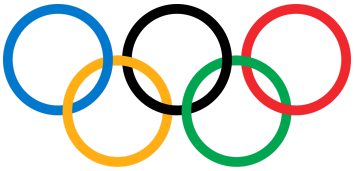
UVA Olympics



UVA has decided to start hosting their own Olympic games! But to be a little quirky and unique and cool and neat and quirky, they have decided that each student who participates has to enter into **EVERY** event. They don't really know what type of athletic events they're going to do yet, nor even how many (this is a very recent idea). But UVA is sure that once the events are done, they want to make sure that the students who did well got credit for their efforts. They consider posting the top 3 winners of each event, but why not give credit to every student? Every student that participates deserves to be recognized (even if they did horrible lol)!

So, they decided on their website they're going to put a massive leaderboard of all the contestants, and the athletes can be ranked and re-ranked by their scores for a selected event, with higher scores meaning better performance. They decided that if two people tied for a given selected event, whoever was originally ranked higher would remain ranked higher. The initial ranking is simply the order the participants were put into the system. Your job, as UVA's only software engineer, is to build this program so that it can sort by specific events and then answer queries as to who came in what place.

Also, It's notable to mention that many types of people, not just athletes, are participating in these games. The people range from being *very* athletic and competitive to students that are just participating for fun. This means that though not always guaranteed, students that do score high in one event generally also do well in other events, and students that score lower in one event aren't exactly the best performers overall. As a result, when UVA re-ranks participants by a new event, the existing ranking is often already pretty close to the new correct order, with long stretches (or *runs*) of participants remaining in nearly the same relative positions.

Input

The first line of input gives the number of students $n \leq 5 * 10^4$, the number of events $m \leq 50$, the next n lines will each give that student's m scores, each separated by a space. Each of these scores are represented with an integer.

The following line will contain the number $C \leq 5 * 10^4$, the number of commands that will be requested on the data. The next C lines will be in one of the following two formats:

- The letter Q (for query) followed by two integers, i j, queries the participants currently ranked in positions [i, j).
- The letter S (for sort) followed by an integer $i < m$ that is the event that the participants will be sorted by.

Output

For each query, output the **ORIGINAL** indices of the participants in the range query in their ranked order, with the 0th index in the ranked order being the highest ranking participant.

Sample Input

```
8 6
1 2 1 2 1 2
4 1 3 4 2 3
1 1 1 0 1 2
9 12 4 8 7 10
5 6 5 5 4 7
4 5 6 7 3 3
8 12 9 9 7 8
4 4 3 6 9 5
5
S 4
Q 1 2
Q 0 8
S 1
Q 0 8
```

Sample Output

```
3
7 3 6 4 5 1 0 2
3 6 4 5 7 0 1 2
```